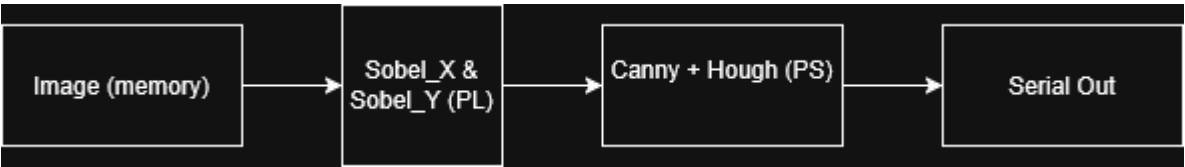


Image Processing Tic-Tac-Toe

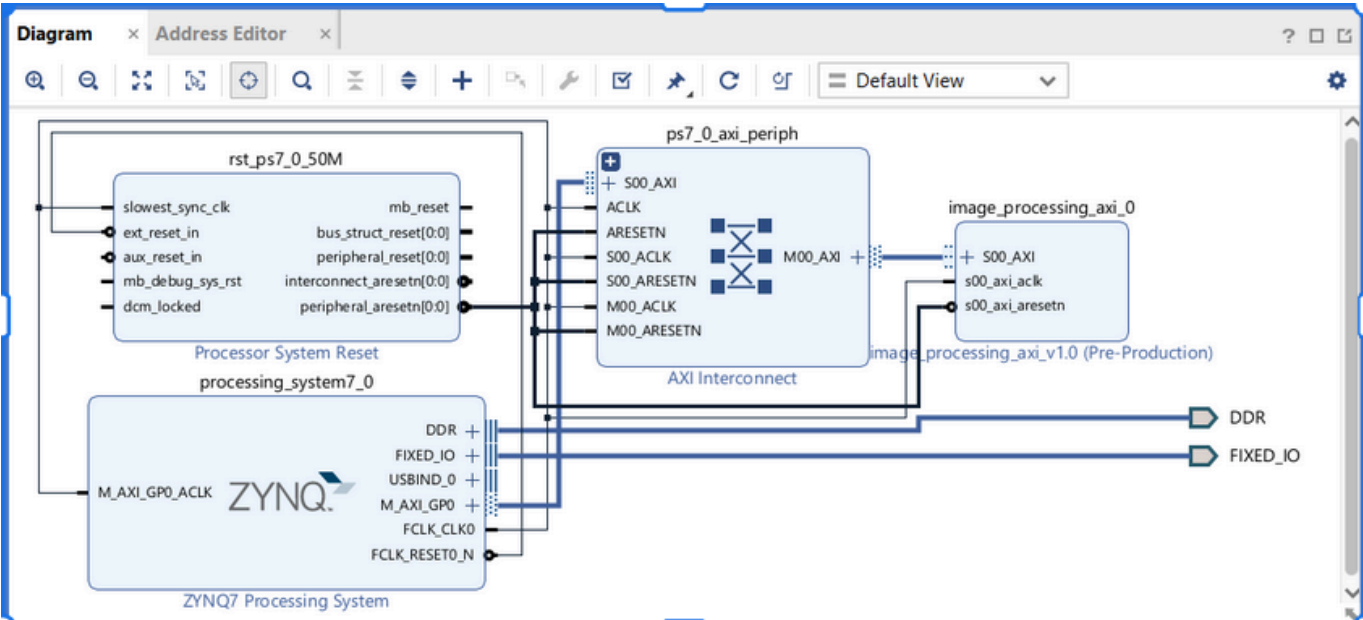
Overview

- Our goal was to create an image processing pipeline that can recognize a game of tic-tac-toe. The project will make use of the Zybo board to hardware-accelerate a part of the image processing pipeline, then finish analyzing the image in software.
- Image stored in memory as an array
- AXI Writes to send image data from PS to PL
- PL - image processing module takes advantage of hardware abilities to quickly perform kernel convolution (sobel x, sobel y)
- PS finishes image processing and uses Hough transform to detect X's vs O's

System Architecture



Hardware Design (PL)



Register Map

Offset	Register	Description
0x00(slv_reg0)	REG_FILTER_LO	Bits [31:0] of the kernel

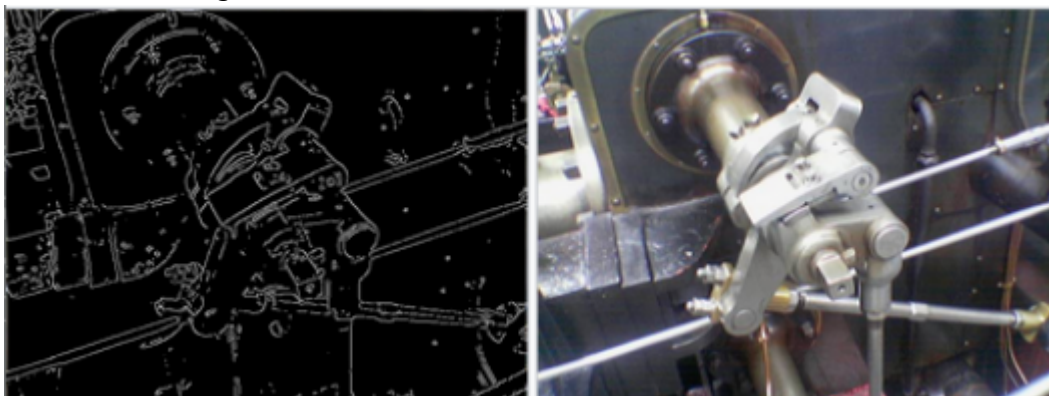
Offset	Register	Description
0x04(slv_reg1)	REG_FILTER_MID	Bits [63:32] of the kernel
0x08(slv_reg2)	REG_FILTER_HI	Bits [71:64] of the kernel at [7:0]
		Scale at [15:8]
		Bias at [24:16]
0x0C(slv_reg3)	REG_IMG_WIDTH	image width
0x10(slv_reg4)	REG_CONTROL	pix_in at [7:0]
		pix_wr at [8]
		reset at [9]
0x14(slv_reg5)	REG_OUTPUT	pix_out at [7:0]
		pix_valid at [8]
		overflow at [9]

Software Design (PS)

Canny Edge Detection

Image Processing technique to reduce noise and get thin, clean edges instead of thick edges Steps:

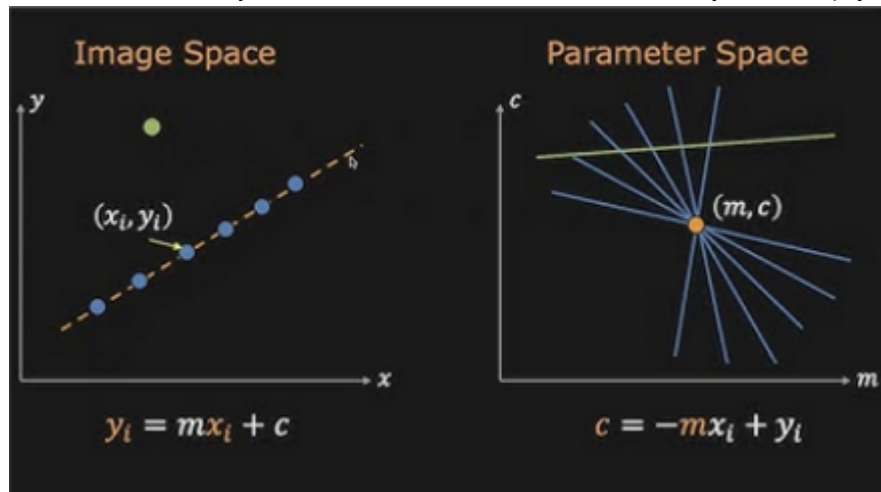
1. Blur
2. Sobel Filter
3. Non-maxima-suppression (turn thick lines into thin lines)
4. Dual-Thresholding



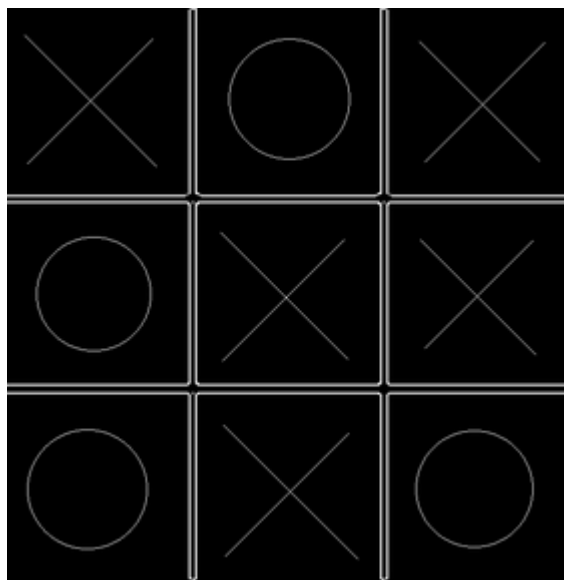
Hough Transform

- Used to detect straight lines and circles in images
- Convert pixels from coordinate space to parameter space ($y = mx + b \rightarrow b = -mx + y$)

- In our case, we only used it to detect X's. We assumed any non-empty space that wasn't an X was an O.



Results



```
Vitis Serial Terminal
Connected to: Serial ( COM7, 115200, 0, 8 )

Connected to COM7 at 115200
Image Processing: Tic-Tac-Toe Recognition
Running Sobel_X...Done
Running Sobel_Y...Done
Combining Sobel_X and Sobel_Y into one image...Done
Scanline analysis...Done
Found peaks at columns 65, 134, and rows 50, 100
Classifying Cells...Done
Game board state:
X|O|X
--+--
O|X|X
--+--
O|X|O
Done
```